JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY GURAJADA VIZIANAGARAM

SRI SIVANI COLLEGE OF ENGINEERING (AFFILIATED TO JNTU-GV)

DEPARTMENT OF CIVIL ENGINEERING

★ OFFICIAL ACADEMIC SYLLABUS & BLUEPRINT ★

B.Tech Civil Engineering R23 Surveying & Geomatics Syllabus Blueprint

Course Code:	R23-CIV-1103	**Academic Scheme:**	I B.Tech I Semester (R23 Regulation)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Unit I: Introduction to Surveying & Compass Traverse

Principles of surveying, Classifications, Errors and corrections in chaining/taping, Prismatic and surveyor compass, Local attraction and declination correction.

Unit II: Levelling & Contouring

Dumpy and auto levels, Staff readings, Height of Instrument (HI) vs Rise and Fall methods, Curvature and refraction corrections, Contour characteristics and interpolation methods.

Unit III: Theodolite Surveying & Trigonometrical Levelling

Vernier transit theodolite, Temporary and permanent adjustments, Measurement of horizontal and vertical angles, Prolonging a straight line, Tachometric surveying principles.

Unit IV: Curves & Setting Out Works

Elements of simple, compound, and reverse circular curves, Setting out simple curves by Rankine's deflection angle method, Transition curves and vertical curves.

Unit V: Modern Geomatics & Total Station / GPS

Principles of Electronic Distance Measurement (EDM), Total Station features and field data handling, Global Positioning System (GPS) segments, Introduction to GIS spatial layers.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Internal Evaluation	30 Marks
University Examination	70 Marks
Field Practical Component	1.5 Credits (3 Hours/Week)

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM